KARAN PRATAP SINGH RATHORE

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EDUCATION

Year	Degree/Certificate & Institute/Board	Institute	CGPA/Marks
2022–2026	B. Tech Indian Institute of Technology Jodhpur	IIT Jodhpur	8.58 / 10
2020–2021	Senior Secondary Examination RBSE	Vyas Public Senior Sec. School Bikaner	99.40%

WORK EXPERIENCE

Reliance Industries Limited – Summer Intern *May 2025 – July 2025*

- Built a predictive maintenance pipeline in Python, processing six months of industrial sensor data to forecast equipment failures.
- Engineered feature-extraction and data-processing workflows for time-series analysis, validating outputs against engineering simulations.

IIChE, IIT Jodhpur – Developer *August 2023 – November 2024*

- Designed and deployed the official chapter website using HTML/CSS/JS and the React framework, achieving a 50-point improvement in Google Lighthouse performance.
- Led UI/UX implementation and feature rollouts for chapter events, delivering a responsive, mobile-friendly platform that increased student engagement.
- Onboarded and mentored junior chapter developers on React component design and Git workflows to support ongoing site maintenance.

PROJECTS

PersonaShift – AI-Powered Interview & Career Simulation Platform *July 2026*

- Architected a full-stack AI interview simulation platform featuring real-time transcript streaming, adaptive interviewer personas, speech-based interactions, and post-session performance analytics.
- Implemented secure authentication and user management with Google OAuth 2.0, email OTP verification, JWT-based authorization, bcrypt password hashing, and Supabase PostgreSQL with role-based access control.
- Developed an asynchronous ATS resume-parsing and semantic evaluation pipeline using BullMQ, Redis, pdf-parse, and local transformer embedding models to enable scalable background processing and automated resume analysis.

AI Dev Team – Autonomous Multi-Agent Software Generation Platform *August 2026*

- Architected a multi-agent orchestration pipeline using LangGraph, Google Gemini, and Docker to autonomously convert natural-language requirements into deployed, full-stack applications end-to-end.
- Designed a 5-stage architecture agent and deterministic cross-validator performing schema, API, and dependency integrity checks with zero LLM cost, eliminating false-positive validation loops and cutting redundant model calls by 60%.
- Built a self-correcting code generation loop (Coder → Reviewer → Executor → Debugger) with tiered failure escalation and git-based rollback, enabling autonomous bug resolution across generated projects without human intervention.

CodeBulb – Copilot for VS Code *December 2025 – January 2026*

- Integrated Google Gemini and DeepSeek APIs into a VS Code extension to enable real-time code generation, debugging, and project scaffolding.
- Developed autonomous AI agents for file management, shell execution, and code generation using Node.js, with a custom WebView interface.
- Tested and debugged extension features across multiple project types, then published the extension on the VS Code Marketplace.

Fisher–Yates Algorithm Visualizer *November 2025*

- Created an interactive single-page application to visualize the Fisher–Yates shuffle algorithm using asynchronous JavaScript animations and DOM manipulation.
- Implemented performance reporting with jsPDF and custom regex-based sanitization to generate downloadable execution reports.
- Optimized DOM update logic to maintain smooth animation performance throughout the shuffle visualization.

COMPETITIONS / CONFERENCES

Amazon ML Challenge 2025 – Smart Product Pricing *2025*

- Placed in the top 1% in the Smart Product Pricing Challenge (Amazon ML).
- Engineered a multimodal pricing model combining text embeddings, CLIP-based image features, and metadata, improving SMAPE from a 61–66% baseline to 45–35%, a 20 percentage-point reduction.

TECHNICAL SKILLS

- **Programming:** JavaScript (ES6+), TypeScript, Python, C++, SQL
- **Frontend:** React.js, HTML5, CSS3, Bootstrap, Tailwind CSS
- **Backend:** Node.js, Express.js, REST APIs, FastAPI, Backend System Design, API Design, Scalable Backend Development, Authentication
- **AI Engineering:** Generative AI, LLM Applications, AI Agents, RAG, GraphRAG, Function Calling, Prompt Engineering, Semantic Search, Knowledge Graphs, Vector Databases, Embedding Models, LangChain, LangGraph, NumPy, Pandas
- **Databases:** PostgreSQL, MySQL, Vector DB (Pinecone), Graph DB (Neo4j)
- **Developer Tools:** Git, GitHub, Linux, Docker, Postman
- **Software Engineering:** Agile Development, Object-Oriented Design, API Design, Performance Optimization, Code Refactoring

RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Probability & Statistics, Scientific Computations, Process Control & AI Applications.

POSITIONS OF RESPONSIBILITY

- **Coordinator:** Career Guidance Cell, IIT Jodhpur *May 2024 – Apr 2025*
- **Developer Head:** IIChE Student Chapter, IIT Jodhpur *Sep 2024 – Dec 2024*

AWARDS AND ACHIEVEMENTS

- Graduated with Department Rank 2, B.Tech in Chemical Engineering, IIT Jodhpur, 2026.